

UNIX Command

Basic Frequently Used Commands

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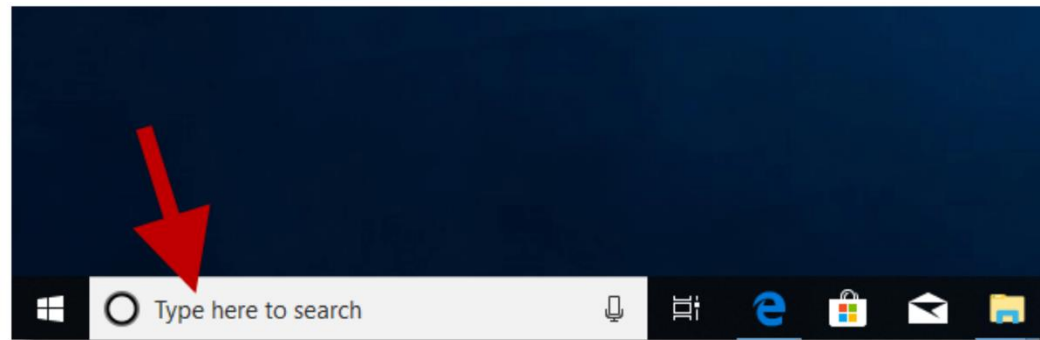
Example of a Unix/Linux **command line interface** (shell)

```
jjm@John-Desk2: ~/demo
1 #!/bin/bash
2 #
3 echo
4 echo "You invoked the script using the name $0"
5 echo "There were $# arguments. They are: " $@
6 echo "The first two args were: \"$1\" and \"$2\""
7 echo
8 echo "The complete list of args one per line (using \"$@\" is:"
9 for arg in "$@"
10 do
11     echo "\"${arg}\""
12 done
13 echo
14 echo "Using \"$@\" without the quotes doesn't do the same thing:"
15 for arg in $@
16 do
17     echo "\"${arg}\""
18 done
19 echo
20 echo "It acts as if you'd used \"$*"
21 for arg in $*
22 do
23     echo "\"${arg}\""
24 done
25 echo
26 echo "But not as if you'd used \"$*\"!"
```

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Accessing your local terminal or command prompt

- **Mac OS**
- Press *command + spacebar* -> *Spotlight Search* -> type *Terminal*
- **Windows 10** or greater
- type *cmd* in the search box (lower left), then choose the highlighted *Command Prompt*



What is Unix?

- Unix is another proprietary ***operating system (OS)***
 - like Windows or MacOS (Unix based)
- There are many Unix and Unix-like operating systems, including open source Linux and its multiple distributions
- To work on the ***command line***, we need to be able to issue ***Unix commands*** to tell the computer what we want it to do

Ubuntu, Debian, AWS

cd -> change directory

How much Unix do we need to learn?

- **Navigating the File System:** Understanding the *hierarchical structure* of *file directories*, using the **cd** command to move between directories.
- **File Paths:** Learning how to specify the location of files using **absolute** and *relative paths*.
- **Basic Unix Commands:** Getting used with common commands like:
 - **ls** for listing files,
 - **mv** for moving files,
 - **rm** for removing files,
 - **mkdir** for creating directories,
 - **cat** for viewing file contents
- **Command Customization:** Learning how to modify the behaviour of Unix commands using **ls -l** to list files with detailed information

absolute path -> full path

File & Directory Management

Task	Command
Print working directory <i>Where are you now</i>	<code>pwd</code> → <i>tell u your local</i> <code>/home/meria_dev</code>
Change directory	<code>cd folder</code>
List files	<code>ls</code>
Long list (permissions) <i>rw-rw-rw-r</i>	<code>ls -l</code>
Show hidden files	<code>ls -a</code>
Up one level (<i>back to the previous directory</i>)	<code>cd ..</code>
Home directory	<code>cd ~</code> <i>relative path - reduce typing process</i>
Make directory	<code>mkdir folder</code>
Remove empty directory	<code>rmdir folder</code>
Remove file	<code>rm file</code>
Remove folder (<i>recursive -> forcing it</i>)	<code>rm -r folder</code>
Copy file	<code>cp file1 file2</code>
Copy folder	<code>cp -r folder1 folder2</code>
Move / rename	<code>mv old new</code>

Be careful: 'rm -r' will permanently delete your folder

vi trial.txt open file to write
i insert
ESC + :X! to save
ESC + .q! quit w/o saving

View File Contents

Task		Command
View entire file		<code>cat file.txt</code> <i>view content of file - what has been written</i>
Scroll through file	<i>type "q" to quit the view</i>	<code>less file.txt</code>
First 10 lines		<code>head file.txt</code>
Last 10 lines		<code>tail file.txt</code>

*What if i want to view first 20 lines?
- head -n 20 filename.txt*

Permissions

chmod -> change mode

Task	Command
Change permission	<code>chmod 755 file</code>
View permissions	<code>ls -l file</code>

d -> directory / folder
- -> file

File Permissions

chmod *octal file* - change the permissions of *file* to *octal*, which can be found separately for **user**, **group**, and **world** by adding:

- 4 - read (r)
- 2 - write (w)
- 1 - execute (x)

Examples:

chmod 777 - read, write, execute for all

chmod 755 - rwx for owner, rx for group and world

	user	group	world			
-rw-r--r--@	1	bernardlee	staff	190358		
-rw-r--r--@	1	bernardlee	staff	2451463		
-rw-r--r--@	1	bernardlee	staff	926889		
-rw-r--r--@	1	bernardlee	staff	1004183		
-rw-r--r--@	1	bernardlee	staff	7789		
-rw-r--r--	1	bernardlee	staff	31		
-rw-r--r--@	1	bernardlee	staff	165		
-rw-r--r--@	1	bernardlee	staff	165		
6	4	4				

Search Commands

Task	Command
Search text inside files	<code>grep "text" file</code>
Recursive search (search <i>inside all files in this folder and its subfolders</i>)	<code>grep -r "text" .</code>
Find files by name	<code>find . -name "*.txt"</code>

Notes:

1. the **dot** (.) -> refers to *this current folder*
2. **asterisk** (*) -> *wild card character* to represent everything

if you want to kill the job

System Commands

Task	Command
Who am I?	whoami
System info	uname -a
Running processes <i>to quit -> type 'q'</i>	top
Disk usage	df -h
Folder size	du -sh folder
Command history	history

we want to copy all the command that we executed before?

- history > all_commands.txt

- history | grep "head -n"

“|” -> this is called pipe -> feed forward

Network Commands

To stop the ping process -> type 'ctr + c'

Task	Command
Ping a server	ping google.com.my
Download/request URL	curl https://example.com

curl --version

Example unix code:

Download and keep original filename (capital O)

1. **curl -O** https://www.gnu.org/licenses/gpl-3.0.txt

If curl is not installed

- sudo apt update

DOWNLOAD & RENAME FILE (small o)

2. **curl -o** gpl_page.html https://www.gnu.org/licenses/gpl-3.0.txt

- sudo apt install curl

DOWNLOAD WITH PROGRESS BAR

3. **curl -# -O** https://www.gnu.org/licenses/gpl-3.0.txt

4. **wc -l** gpl-3.0.txt

wc -> word count

-l -> how many lines

Zip & Tar

Task	Command
Create tar.gz	<code>tar -czf file.tar.gz folder/</code>
Extract tar.gz	<code>tar -xzf file.tar.gz</code>
Zip folder	<code>zip -r file.zip folder/</code>
Unzip	<code>unzip file.zip</code>

Side-by-Side Comparison

Feature	<code>tar -czf</code>	<code>zip</code>
Creates archive	✓ Yes	✓ Yes
Compression	gzip	zip
Default on Linux	✓ Yes	✗ No
Windows compatible	✗ Needs tool	✓ Native
Preserves permissions	✓ Yes	⚠ Limited
Common file	<code>.tar.gz</code>	<code>.zip</code>

Other Useful Commands

Task	Command
Print text	echo "hello"
Clear terminal	clear

Example unix code:

echo -> display

1. name=Bernard
2. echo \$name

Write text into a file

3. echo "Unix is fun" > file.txt

“>” means output the file

Append text to a file

4. echo "Hello world" >> file.txt
5. cat file.txt

“>>” means append / sambung

For more Unix Command

- <https://tinyurl.com/5n6mwwsk>

Thank you!!!